LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Precision measurements, characteristic of precision measuring instruments.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Precision measurements, characteristic of precision measuring instruments.
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of precision instrument handling; Safety: Handle glass/metal parts gently; Supervision: Instructor monitors grip and alignment.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engineering Metrology?
(b) Name one common unit of length used in aviation engineering.
(c) Why is precision critical in aircraft component manufacturing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Engineering Metrology • Importance of precision in aviation • Sources of measurement error • Role of standards in metrology • Difference between accuracy and precision	L & Ds	PPT, IB	15 min
(b)	• Characteristics of precision instruments • Sensitivity and resolution concepts • Stability and repeatability factors • Environmental effects on measurements • Calibration requirements for instruments	L & Ds	PPT, PDF	20 min
(c)	• Selection criteria for precision tools • Maintenance and storage protocols • Common defects in measuring instruments • Troubleshooting measurement discrepancies • Documentation of calibration records	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between accuracy and precision in your own words.
(b) List three environmental factors that affect precision measurements.
(c) Why is regular calibration essential for precision instruments?

12. Home Assignment and Follow Up	Prepare a brief note on the importance of Engineering Metrology in the context of the next lesson on Vernier instruments (LP 04).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Vernier instruments, reading of Vernier scale, types of Vernier Caliper, precautions while using the instrument.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min